

61. Let $L_1 : \frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0}$ and $L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{\alpha}$, $\alpha \in \mathbb{R}$ be two lines, which intersect at the point B. If P is the foot of perpendicular from the point A(1, 1, -1) on L_2 , then the value of $26 \alpha(PB)^2$ is _____.

माना कि $L_1 : \frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0}$ और $L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{\alpha}$, $\alpha \in \mathbb{R}$, दो रेखाएँ हैं, जो कि बिन्दु B पर प्रतिच्छेद करती हैं।

यदि P, बिन्दु A(1, 1, -1) से L_2 पर लम्ब का पाद है, तो $26 \alpha(PB)^2$ का मान है _____.

Ans. 216

Sol. Point B

$$(3\lambda + 1, -\lambda + 1, -1) \equiv (2\mu + 2, 0, \alpha\mu - 4)$$

$$3\lambda + 1 = 2\mu + 2$$

$$-\lambda + 1 = 0$$

$$-1 = \alpha\mu - 4$$

$$\lambda = 1, \mu = 1, \alpha = 3$$

$$B(4, 0, -1)$$

$$\text{Let Point 'P' is } (2\delta + 2, 0, 3\delta - 4)$$

$$\text{Dir's of AP} < 2\delta + 1, -1, 3\delta - 3 >$$

$$AP \perp L_2 \Rightarrow \delta = \frac{7}{13}$$

$$P\left(\frac{40}{13}, 0, \frac{-31}{13}\right)$$

$$26\alpha(PB)^2 = 26 \times 3 \times \left(\frac{144}{169} + \frac{324}{169}\right)$$

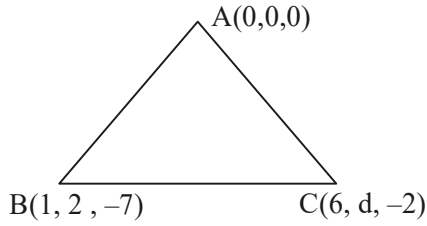
$$= 216$$

62. Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\overrightarrow{AB} = \hat{i} + 2\hat{j} - 7\hat{k}$, $\overrightarrow{BC} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\overrightarrow{AC} = 6\hat{i} + d\hat{j} - 2\hat{k}$, $d > 0$. Then the square of the length of the largest side of the triangle ABC is _____.

माना एक त्रिभुज ABC का क्षेत्रफल $15\sqrt{2}$ है तथा सदिश $\overrightarrow{AB} = \hat{i} + 2\hat{j} - 7\hat{k}$, $\overrightarrow{BC} = a\hat{i} + b\hat{j} + c\hat{k}$ और $\overrightarrow{AC} = 6\hat{i} + d\hat{j} - 2\hat{k}$, $d > 0$ हैं। तो त्रिभुज ABC की सबसे बड़ी भुजा की लम्बाई का वर्ग है _____

Ans. 54

Sol.



$$\frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -7 \\ 6 & d & -2 \end{vmatrix} = \pm 30\sqrt{2}$$

$$\hat{i}(-4+7d) - \hat{j}(-2+42) + \hat{k}(d-12) = \pm 30\sqrt{2}$$

$$\Rightarrow (7d-4)^2 + 40^2 + (d-12)^2 = 1800$$

$$49d^2 + 16 - 56d + 1600 + d^2 + 144 - 24d = 1800$$

$$50d^2 - 80d - 40 = 0$$

$$5d^2 - 8d - 4 = 0$$

$$5d^2 - 10d + 2d - 4 = 0$$

$$(5d+2)(d-2) = 0$$

$$d = 2$$

$$AB = \sqrt{54}$$

$$AC = \sqrt{36+4+4} = \sqrt{44}$$

$$BC = \sqrt{25+0+25}$$

63. Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times \vec{c} = \hat{i} + 8\hat{j} + 13\hat{k}$. If $\vec{a} \cdot \vec{c} = 13$, then $(24 - \vec{b} \cdot \vec{c})$ is equal to _____.

माना $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ तथा सदिश $\vec{c}$ इस प्रकार है कि $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times \vec{c} = \hat{i} + 8\hat{j} + 13\hat{k}$ है। यदि $\vec{a} \cdot \vec{c} = 13$, तो $(24 - \vec{b} \cdot \vec{c})$ बराबर है _____

Ans. 46

Sol. $\vec{a} \times \vec{b} + \vec{a} \times \vec{c} + \vec{b} \times \vec{c} = (1, 8, 13)$

$$\vec{a} \times (\vec{a} \times \vec{b}) + \vec{a} \times (\vec{a} \times \vec{c}) + \vec{a} \times (\vec{b} \times \vec{c})$$

$$= \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$$

$$(\vec{a} \cdot \vec{b})\vec{a} - a^2\vec{b} + (\vec{a} \cdot \vec{c})\vec{a} - a^2\vec{c} + (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$$

$$\Rightarrow -26\vec{a} - 29\vec{b} + 13\vec{a} - 29\vec{c} + 13\vec{b} + 26\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$$

$$\Rightarrow -13\vec{a} - 16\vec{b} - 3\vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k})$$

$$\Rightarrow -13\vec{a} \cdot \vec{b} - 16\vec{b} \cdot \vec{c} - 3\vec{b} \cdot \vec{c} = \vec{a} \times (\hat{i} + 8\hat{j} + 13\hat{k}) \cdot \vec{b}$$

$$\Rightarrow (-13)(-26) - 16(50) - 3\vec{b} \cdot \vec{c} = \begin{vmatrix} 2 & -3 & 4 \\ 1 & 8 & 13 \\ 3 & 4 & -5 \end{vmatrix}$$

$$\Rightarrow -462 - 3\vec{b} \cdot \vec{c} = -396$$

$$\Rightarrow \vec{b} \cdot \vec{c} = -22$$

$$\text{Hence } 24 - \vec{b} \cdot \vec{c} = 46$$

64. Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that

$$\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a} \text{ and } \vec{r} \cdot (\vec{b} - \vec{c}) = 0, \text{ then } \frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2} \text{ is equal to } \underline{\hspace{2cm}}.$$

माना $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ तथा $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ तीन सदिश दिये गये हैं यदि $\vec{r}$ एक सदिश इस प्रकार है

$$\text{कि } \vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a} \text{ तथा } \vec{r} \cdot (\vec{b} - \vec{c}) = 0 \text{ है, तो } \frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2} \text{ बराबर होगा } \underline{\hspace{2cm}}$$

Ans. 569

Sol. $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$

$$\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$$

$$\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$$

$$(\vec{r} - (\vec{b} + \vec{c})) \times \vec{a} = 0$$

$$\Rightarrow \vec{r} = (\vec{b} + \vec{c}) + \lambda \vec{a}$$

$$\vec{r} = (20\hat{i} + 5\hat{j} - 12\hat{k}) + \lambda(9\hat{i} - 13\hat{j} + 25\hat{k})$$

$$= (20 + 9\lambda)\hat{i} + (5 - 13\lambda)\hat{j} + (25\lambda - 12)\hat{k}$$

$$\text{Now, } \vec{r} \cdot (\vec{b} - \vec{c}) = 0$$

$$\vec{r} \cdot (-14\hat{i} + 9\hat{j} - 14\hat{k}) = 0$$

$$\text{Now } -14(20 + 9\lambda) + 9(5 - 13\lambda) - 14(25\lambda - 12) = 0$$

$$-593\lambda - 67 = 0$$

$$\lambda = -\frac{67}{593}$$

$$\therefore \vec{r} = (\vec{b} + \vec{c}) - \frac{67}{593} \vec{a}$$

$$\frac{|593\vec{r} + 67\vec{a}|^2}{|593|^2} = |\vec{b} + \vec{c}|^2 = |20\hat{i} + 5\hat{j} - 12\hat{k}|^2$$

$$= 569$$

65. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$, and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ and the angle between $\vec{b}$ and $\vec{c}$ is α , then $192 \sin^2 \alpha$ is equal to _____.

माना दो सदिशों $\vec{a}$ तथा $\vec{b}$ के लिए $|\vec{a}| = 1$, $|\vec{b}| = 4$, तथा $\vec{a} \cdot \vec{b} = 2$ हैं। यदि $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ है तथा $\vec{b}$ और $\vec{c}$ है के बीच कोण α है, तो $192 \sin^2 \alpha$ बराबर है _____.

Ans. 48

Sol. $\vec{b} \cdot \vec{c} = (2\vec{a} \times \vec{b}) \cdot \vec{b} - 3|\vec{b}|^2$

$$|\vec{b}||\vec{c}|\cos \alpha = -3|\vec{b}|^2$$

$$|\vec{c}|\cos \alpha = -12, \text{ as } |\vec{b}| = 4$$

$$\vec{a} \cdot \vec{b} = 2$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

$$|\vec{c}|^2 = |(2\vec{a} \times \vec{b}) - 3\vec{b}|^2$$

$$= 64 \times \frac{3}{4} + 144 = 192$$

$$|\vec{c}|^2 \cos^2 \alpha = 144$$

$$192 \cos^2 \alpha = 144$$

$$192 \sin^2 \alpha = 48$$

66. Let $\vec{a} = 3\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + \vec{b}) \times \vec{c} = 2(\vec{a} \times \vec{b}) + 24\hat{j} - 6\hat{k}$ and $(\vec{a} - \vec{b} + \hat{i}) \cdot \vec{c} = -3$. Then $|\vec{c}|^2$ is equal to _____.

माना $\vec{a} = 3\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$ तथा सदिश $\vec{c}$ इस प्रकार हैं कि $(\vec{a} + \vec{b}) \times \vec{c} = 2(\vec{a} \times \vec{b}) + 24\hat{j} - 6\hat{k}$ तथा $(\vec{a} - \vec{b} + \hat{i}) \cdot \vec{c} = -3$ हैं। तो $|\vec{c}|^2$ बराबर है _____

Ans. 38

Sol. $(\vec{a} + \vec{b}) \times \vec{c} = 2(\vec{a} \times \vec{b}) + 24\hat{j} - 6\hat{k}$
 $(5\hat{i} + \hat{j} + 4\hat{k}) \times \vec{c} = 2(7\hat{i} - 7\hat{j} - 7\hat{k}) + 24\hat{j} - 6\hat{k}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 1 & 4 \\ x & y & z \end{vmatrix} = 14\hat{i} + 10\hat{j} - 20\hat{k}$$

$$\Rightarrow \hat{i}(z - 4y) - \hat{j}(5z - 4x) + \hat{k}(5y - x) = 14\hat{i} + 10\hat{j} - 20\hat{k}$$

$$z - 4y = 14, 4x - 5z = 10, 5y - x = -20$$

$$(\vec{a} - \vec{b} + \hat{i}) \cdot \vec{c} = -3$$

$$(2\hat{i} + 3\hat{j} - 2\hat{k}) \cdot \vec{c} = -3$$

$$2x + 3y - 2z = -3$$

$$\therefore x = 5, y = -3, z = 2$$

$$|\vec{c}|^2 = 25 + 9 + 4 = 38$$

67. Let a line passing through the point $(-1, 2, 3)$ intersect the lines $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ at $M(\alpha, \beta, \gamma)$ and

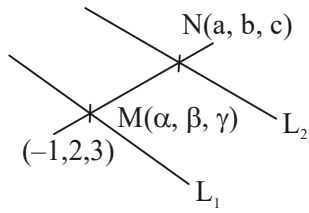
$$L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4} \text{ at } N(a, b, c). \text{ Then the value of } \frac{(\alpha + \beta + \gamma)^2}{(a + b + c)^2} \text{ equal } \underline{\hspace{2cm}}.$$

माना बिन्दु $(-1, 2, 3)$ से होकर जाने वाली रेखा, रेखा $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ को $M(\alpha, \beta, \gamma)$ तथा रेखा

$$L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4} \text{ को } N(a, b, c) \text{ पर काटती है। तो } \frac{(\alpha + \beta + \gamma)^2}{(a + b + c)^2} \text{ बराबर है } \underline{\hspace{2cm}}$$

Ans. 196

Sol.



$$M(3\lambda + 1, 2\lambda + 2, -2\lambda - 1)$$

$$N(-3\mu - 2, -2\mu + 2, 4\mu + 1)$$

$$\frac{3\lambda + 2}{-3\mu - 1} = \frac{2\lambda}{-2\mu} = \frac{-2\lambda - 4}{4\mu - 2} \Rightarrow \mu = 2, \lambda = 4$$

$$\text{So, } \begin{matrix} \alpha + \beta + \gamma = 14 \\ a + b + c = -1 \end{matrix} \Rightarrow \frac{(\alpha + \beta + \gamma)^2}{(a + b + c)^2} = 196$$

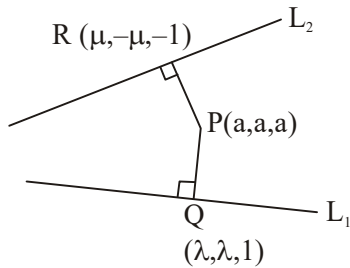
68. Let Q and R be the feet of perpendiculars from the point (a, a, a) on the lines $x = y, z = 1$ and $x = -y, z = -1$ respectively. If $\angle QPR$ is a right angle, then $12a^2$ is equal to _____.

माना बिन्दु $P(a, a, a)$ से रेखाओं $x = y, z = 1$ तथा $x = -y, z = -1$ पर डाले गए लंबों के पाद क्रमशः Q तथा R हैं। यदि $\angle QPR$ एक समकोण है, तो $12a^2$ बराबर है _____.

Ans. 12

Sol. $L_1 : \frac{x}{1} = \frac{y}{1} = \frac{z-1}{0} = \lambda$ (let)

$$L_2 : \frac{x}{1} = \frac{y}{-1} = \frac{z+1}{0} = \mu$$
 (let)



$$PQ \perp L_1$$

$$(\lambda - a)1 + (\lambda - a)1 + (1 - a)0 = 0$$

$$\lambda = a$$

$$PR \perp L_2$$

$$1(\mu - a) + 1(\mu + a) + 0(-1 - a) = 0$$

$$\mu - a + \mu + a = 0$$

$$\mu = 0$$

$$\therefore \text{Point } Q(a, a, 1) \text{ and } R(0, 0, -1)$$

According to question

$$\angle QPR = 90^\circ$$

$$\text{Direction ratio of PR} = (a, a, a + 1)$$

$$\text{Direction ratio of PQ} = (0, 0, a - 1)$$

$$a \times 0 + 0 \times a + (a + 1)(a - 1) = 0$$

$$a^2 = 1$$

$$\Rightarrow 12a^2 = 12 \times 1 = 12$$

69. A line passes through A (4, -6, -2) and B (16, -2, 4). Then point P (a, b, c), where a, b, c are non-negative integers, on the line AB lies at a distance of 21 units, from the point A. The distance between the points P(a, b, c) and Q(4, -12, 3) is equal to _____.

एक रेखा, बिन्दुओं A (4, -6, -2) तथा B (16, -2, 4) से होकर जाती है। रेखा AB पर बिन्दु P(a, b, c), जहाँ a, b, c ऋणोत्तर पूर्णांक हैं, बिन्दु A से 21 इकाई की दूरी पर है। बिन्दुओं P(a, b, c) तथा Q(4, -12, 3) के बीच दूरी है _____

Ans. 22

Sol. A(4 - 6 - 2) B(16 - 2, 4)

Equation of line AB :

$$\frac{x - 4}{6} = \frac{y + 6}{2} = \frac{z + 2}{3}$$

$$\frac{x - 4}{6} = \frac{y + 6}{2} = \frac{z + 2}{3} = 21$$

$$x = 22, \quad y = 0, \quad z = 7$$

$$P(22, 0, 7), \quad Q(4, -12, 3)$$

$$PQ = \sqrt{(18)^2 + (12)^2 + (4)^2}$$

$$= \sqrt{484} = 22$$